Residual Estimator and Reliability

Element and facet residuals, local indicators, and an energy-error bound

Lecture outline

We begin with the elementwise integration-by-parts formula obtained in the previous lecture.

1. Regroup element-boundary terms into interior, Neumann, and Dirichlet facet contributions.
2. Compute element and facet residuals in one and two dimensions.
3. Derive the mesh-size weights in the residual estimator.
4. Prove that the estimator gives a reliable upper bound for the energy error.

The two worked examples show where the element and facet terms enter. The reliability proof then explains why these computable terms control the unknown error.

Review: residual and Galerkin orthogonality

For the diffusion problem,

$$R(u_h; v) = \ell(v) - a(u_h, v), \qquad v \in \mathcal{V}_0.$$

If $e = u - u_h$, then the continuous equation gives

$$R(u_h; v) = a(e, v), \qquad \|e\|_a^2 = R(u_h; e).$$

The finite element equation gives

$$R(u_h; v_h) = 0 \qquad \forall v_h \in \mathcal{V}_{h,0}.$$

Thus the residual measures the part of the continuous problem not tested by the discrete space. Galerkin orthogonality will later allow us to replace an unknown test function by its finite element approximation error.

Review: integration by parts on each element

The previous lecture stopped after applying Green's formula separately on every element:

$$\int_{\mathcal{K}_e} k\nabla u_h \cdot \nabla v \, \mathrm{d}x = -\int_{\mathcal{K}_e} \nabla\cdot(k\nabla u_h)v \, \mathrm{d}x + \int_{\partial\mathcal{K}_e} (k\nabla u_h \cdot \boldsymbol{n}_e)v \, \mathrm{d}s.$$

The first term measures the balance defect inside $\mathcal{K}_e$. The second term lies on the element boundary.

The next step is to sum over all elements and determine what the boundary terms become on each class of facet.

After summing, every element boundary must be classified

Substituting the elementwise identity into the residual gives

$$R(u_h; v) = \sum_e \int_{\mathcal{K}_e} \big(f + \nabla\cdot(k\nabla u_h)\big) v \, \mathrm{d}x \\ - \sum_e \int_{\partial\mathcal{K}_e} (k\nabla u_h \cdot \boldsymbol{n}_e) v \, \mathrm{d}s + \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The first line is already separated by element. In the second line:

- an interior facet appears twice, once from each neighboring element;
- a Neumann facet appears once and combines with the prescribed-flux integral; and
- a Dirichlet facet contributes zero because $v = 0$ on Γ_D.

Interior facet: the two outward normals have opposite signs

Let $\mathcal{F} = \partial\mathcal{K}^+ \cap \partial\mathcal{K}^-$. The outward normals are $\boldsymbol{n}_\mathcal{F}$ for $\mathcal{K}^+$ and $-\boldsymbol{n}_\mathcal{F}$ for $\mathcal{K}^-$. The two boundary terms are

$$
\begin{aligned}
&- \int_\mathcal{F} k \nabla u_h^+ \cdot \boldsymbol{n}_\mathcal{F}\, v \, \mathrm{d}s - \int_\mathcal{F} k \nabla u_h^- \cdot (-\boldsymbol{n}_\mathcal{F})\, v \, \mathrm{d}s \\
&\qquad = - \int_\mathcal{F} \left(k \nabla u_h^+ - k \nabla u_h^- \right) \cdot \boldsymbol{n}_\mathcal{F}\, v \, \mathrm{d}s.
\end{aligned}
$$

Because $v \in H^1(\Omega)$ has the same boundary value on the shared facet when approached from either conforming element, the same v appears in both terms. The coefficient of v in the resulting facet integral is the negative normal-flux jump. Define the interior-facet residual by

$$
j_\mathcal{F} = -[\![k \nabla u_h \cdot \boldsymbol{n}_\mathcal{F}]\!]_\mathcal{F}.
$$

Boundary facets and the complete residual representation

Define the computed vector flux element by element:

$$\boldsymbol{q}_h = k\nabla u_h.$$

On a Neumann facet, the element-boundary term and prescribed-flux term give

$$j_{\mathcal{F}} = q - k\nabla u_h \cdot \boldsymbol{n} = q - \boldsymbol{q}_h \cdot \boldsymbol{n}.$$

On a Dirichlet facet, $v = 0$. Define

$$r_e = f + \nabla\cdot(k\nabla u_h) = f + \nabla\cdot \boldsymbol{q}_h.$$

The residual is therefore

$$\boxed{R(u_h; v) = \sum_e \int_{\mathcal{K}_e} r_e v \, \mathrm{d}x + \sum_{\mathcal{F}\in\mathcal{F}_h^{\mathrm{int}}\cup\mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} v \, \mathrm{d}s.}$$

This is an exact identity, not an estimate.

What the two residuals measure

Element residual
$r_e = f + \nabla \cdot \boldsymbol{q}_h$
balance defect inside $\mathcal{K}_e$

Facet residual
$j_{\mathcal{F}}$
normal-flux jump or
Neumann mismatch

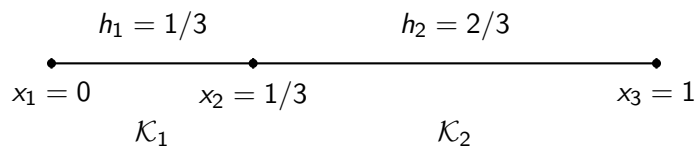
For P_1 elements and elementwise-constant k, ∇u_h and $k\nabla u_h$ are constant on each element. Hence

$$\nabla \cdot (k \nabla u_h) = 0, \qquad r_e = f|_{\mathcal{K}_e}.$$

The element residual then reduces to the prescribed source. The computed-gradient information is contained in the facet residuals.

Nonuniform 1D example: compute the residuals

$$-u'' = 1 \text{ in } (0,1), \qquad u(0) = 0, \qquad u'(1) = 0, \qquad (U_1, U_2, U_3) = \left(0, \frac{5}{18}, \frac{1}{2}\right).$$



$$u_h'|_{\mathcal{K}_1} = \frac{5}{6}, \qquad u_h'|_{\mathcal{K}_2} = \frac{1}{3}, \qquad r_1 = r_2 = 1.$$

In one dimension the facets are nodes. Thus $j_2 = 1/3 - 5/6 = -1/2$ and $j_3 = 0 - 1/3 = -1/3$.

What the 1D example shows

For every test function v with $v(0) = 0$,

$$R(u_h; v) = \int_{\mathcal{K}_1} v \, \mathrm{d}x + \int_{\mathcal{K}_2} v \, \mathrm{d}x - \frac{1}{2} v(x_2) - \frac{1}{3} v(x_3).$$

The residual therefore contains both the element terms and the two facet terms.

Effect of element size.

$$h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} = h_e^3, \qquad \|u - u_h\|^2_{a,\mathcal{K}_e} = \frac{h_e^3}{12}.$$

Because $h_2 = 2h_1$, $\mathcal{K}_2$ contributes eight times as much to both squared quantities.

Special feature of 1D. For $w = e - \mathcal{I}_h e$,

$$w(x_i) = 0$$

at every node. Thus the point-facet terms vanish from the 1D reliability proof even though j_2 and j_3 are nonzero. For $d \geq 2$, the facet terms remain in the estimator.

Constructing a residual estimator

Remove the unknown error from the residual identity, identify the mesh-size weights, and obtain a computable upper bound.

What are we trying to construct?

A **local error indicator** η_e is a computable number associated with element $\mathcal{K}_e$. The indicators are used to rank elements when selecting where the mesh should be refined.
The **global estimator** is

$$\eta = \left(\sum_e \eta_e^2 \right)^{1/2}.$$

The desired result is a **reliability bound**

$$\boxed{\|u - u_h\|_a \leq C_{\mathrm{rel}} \eta,}$$

where C_{rel} is independent of the mesh size.
The residuals r_e and $j_{\mathcal{F}}$ are computable, but the correct powers of h_e and $h_{\mathcal{F}}$ have not yet been established. The derivation determines those powers.

Why the exact residual identity is not computable

Let $e = u - u_h$. Taking $v = e$ in the exact residual representation gives

$$\|e\|_a^2 = \sum_e \int_{\mathcal{K}_e} r_e e \, \mathrm{d}x + \sum_{\mathcal{F} \in \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} e \, \mathrm{d}s.$$

The residuals are known, but e is not. For example,

$$\left| \int_{\mathcal{K}_e} r_e e \, \mathrm{d}x \right| \leq \|r_e\|_{L^2(\mathcal{K}_e)} \|e\|_{L^2(\mathcal{K}_e)}$$

still contains the unknown local error norm. It also supplies no factor of h_e. We need to replace e by a locally small function without changing the value of the residual functional.

The quasi-interpolant $\mathcal{Q}_h$

The error e is known only to belong to $H^1(\Omega)$. In two and three dimensions, such a function need not have nodal point values, so the nodal interpolant cannot be applied to e. A **quasi-interpolation operator**

$$\mathcal{Q}_h : \mathcal{V}_0 \longrightarrow \mathcal{V}_{h,0}$$

constructs a finite element function from local averages rather than nodal values. We use it only in the proof; e and $\mathcal{Q}_h e$ are not computed when the estimator is evaluated.
For an element $\mathcal{K}_e$, let ω_e be the union of the elements sharing a vertex with $\mathcal{K}_e$. For a facet $\mathcal{F}$, define $\omega_{\mathcal{F}}$ similarly. These neighborhoods are called **patches**.

Step 1: use Galerkin orthogonality

For every $v_h \in \mathcal{V}_{h,0}$, the finite element equations give

$$R(u_h; v_h) = 0.$$

Hence, for any $v \in \mathcal{V}_0$ and any $v_h \in \mathcal{V}_{h,0}$,

$$R(u_h; v) = R(u_h; v - v_h).$$

Choose $v_h = \mathcal{Q}_h v$, using the quasi-interpolant defined on the preceding slide. With $v = e$, define

$$w = e - \mathcal{Q}_h e.$$

Then

$$\boxed{\|e\|_a^2 = R(u_h; e) = R(u_h; w).}$$

The residual now acts on an approximation error w, whose local size can be bounded by the local mesh size.

Step 2: local approximation supplies the mesh sizes

Let

$$h_e = \operatorname{diam}(\mathcal{K}_e), \qquad h_{\mathcal{F}} = \operatorname{diam}(\mathcal{F}).$$

For $d \geq 2$, a standard quasi-interpolant satisfies

$$\|e - \mathcal{Q}_h e\|_{L^2(\mathcal{K}_e)} \leq C_Q h_e \, |e|_{H^1(\omega_e)} ,$$
$$\|e - \mathcal{Q}_h e\|_{L^2(\mathcal{F})} \leq C_Q \, h_{\mathcal{F}}^{1/2} \, |e|_{H^1(\omega_{\mathcal{F}})} .$$

The constant C_Q depends on mesh shape regularity, but not on the mesh size. The first estimate supplies one power of h_e; the scaled trace estimate on a facet supplies one half power of $h_{\mathcal{F}}$.
In one dimension, facets are points and are treated separately, as in the preceding example.

The element term produces the weight h_e

Recall $w = e - \mathcal{Q}_h e$. For one element,

$$\left| \int_{\mathcal{K}_e} r_e w \, \mathrm{d}x \right| \leq \|r_e\|_{L^2(\mathcal{K}_e)} \, \|w\|_{L^2(\mathcal{K}_e)}$$
$$\leq C_Q \underbrace{h_e \, \|r_e\|_{L^2(\mathcal{K}_e)}}_{\text{computable weighted element residual}} |e|_{H^1(\omega_e)} .$$

The first inequality is Cauchy–Schwarz on $\mathcal{K}_e$. The second uses the element approximation estimate for $\mathcal{Q}_h$.
After squaring, this weighted residual contributes

$$h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)}$$

to the estimator.

The facet term produces the weight $h_{\mathcal{F}}^{1/2}$

For one interior or Neumann facet,

$$\left| \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \right| \leq \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \, \|w\|_{L^2(\mathcal{F})}$$

$$\leq C_Q \qquad \underbrace{h_{\mathcal{F}}^{1/2} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}}_{\text{computable weighted facet residual}} \qquad |e|_{H^1(\omega_{\mathcal{F}})} .$$

The first inequality is Cauchy–Schwarz on $\mathcal{F}$. The second uses the facet approximation estimate for $\mathcal{Q}_h$.
After squaring, this weighted residual contributes

$$h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2$$

to the estimator. The facet term is part of the estimator; it is not optional.

Step 3: define the global residual estimator

Let

$$\mathcal{F}_h^* = \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N.$$

The preceding two bounds identify the **global residual estimator**

$$\boxed{\eta^2 = \sum_e h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} + \sum_{\mathcal{F} \in \mathcal{F}_h^*} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} \cdot}$$

Every quantity in η is computable from the mesh, the data, and u_h.
This is a definition. It becomes an error estimator only after a bound is proved that compares it with the unknown error.

Local indicators distribute the global estimator

For each element $\mathcal{K}_e$, define

$$\eta_e^2 = h_e^2 \, \|r_e\|_{L^2(\mathcal{K}_e)}^2 + \frac{1}{2} \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{\mathrm{int}}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2 + \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{N}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2 .$$

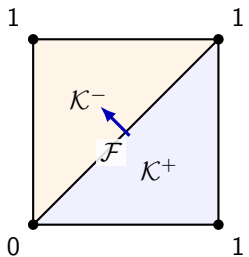
An interior facet is shared by two elements, so one half of its contribution is assigned to each. A Neumann facet belongs to only one element. Therefore

$$\eta^2 = \sum_e \eta_e^2 .$$

The value η_e is an indicator used for marking; it is not the exact error on $\mathcal{K}_e$.

Two triangles: the facet residual carries the defect

Consider $-\Delta u = 0$ on the square, with Dirichlet conditions on the whole boundary. The square is divided along its diagonal, and the nodal values of u_h are shown below.



$$\boldsymbol{n}_{\mathcal{F}} = \frac{(-1,1)}{\sqrt{2}}$$

On each triangle, the P_1 function is determined by its three nodal values:

$$u_h^+(x,y) = x, \qquad u_h^-(x,y) = y.$$

Therefore,

$$\nabla u_h^+ = (1,0), \qquad \nabla u_h^- = (0,1).$$

Because $f = 0$ and these gradients are constant,

$$r_\pm = \nabla \cdot (\nabla u_h^\pm) = 0.$$

The estimator would be zero if it contained only element residuals.

Compute the facet term and the local indicators

Using the normal shown on the previous slide,

$$\begin{aligned} j_{\mathcal{F}} &= -\big(\nabla u_h^+ - \nabla u_h^-\big) \cdot \boldsymbol{n}_{\mathcal{F}} \\ &= -\big((1,0) - (0,1)\big) \cdot \frac{(-1,1)}{\sqrt{2}} = \sqrt{2}. \end{aligned}$$

The diagonal has length $h_{\mathcal{F}} = \sqrt{2}$, so

$$\|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} = \int_{\mathcal{F}} 2 \, \mathrm{d}s = 2\sqrt{2}, \qquad \eta^2 = h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} = 4.$$

The shared-facet contribution is divided equally between its two neighboring elements:

$$\eta_+^2 = \eta_-^2 = 2, \qquad \eta^2 = \eta_+^2 + \eta_-^2.$$

The computed function is continuous, but its gradient is not. The facet residual detects this defect; an element-residual-only estimator would miss it.

Cauchy–Schwarz combines all weighted terms

Substituting $w = e - \mathcal{Q}_h e$ in the residual representation and using the two local bounds gives a sum of products. Cauchy–Schwarz for that finite sum gives

$$\|e\|_a^2 \le C_Q \eta \left(\sum_e |e|^2_{H^1(\omega_e)} + \sum_{\mathcal{F} \in \mathcal{F}_h^*} |e|^2_{H^1(\omega_{\mathcal{F}})} \right)^{1/2}.$$

The first factor is the computable estimator η. The second factor still contains the error, but only through its H^1 seminorm over overlapping patches.

Finite patch overlap gives a global error norm

On a shape-regular mesh, an element belongs to at most a fixed number N_{ov} of the patches ω_e and $\omega_{\mathcal{F}}$. Consequently,

$$\sum_{e} |e|^2_{H^1(\omega_e)} + \sum_{\mathcal{F}\in\mathcal{F}_h^*} |e|^2_{H^1(\omega_{\mathcal{F}})} \le N_{\mathrm{ov}}\, |e|^2_{H^1(\Omega)}\,.$$

This is a counting statement: each element contribution to $|e|^2_{H^1(\Omega)}$ appears in no more than N_{ov} patches. Thus

$$\|e\|_a^2 \le C_Q \sqrt{N_{\mathrm{ov}}}\, \eta\, |e|_{H^1(\Omega)}\,.$$

Reliability of the residual estimator

The diffusion coefficient satisfies $k(\boldsymbol{x}) \geq k_0 > 0$. Hence

$$\|e\|_a^2 = \int_\Omega k|\nabla e|^2 \, \mathrm{d}x \geq k_0 \, |e|_{H^1(\Omega)}^2 \,,$$

and therefore

$$|e|_{H^1(\Omega)} \leq \frac{1}{\sqrt{k_0}} \, \|e\|_a \,.$$

Substituting this inequality into the preceding bound and dividing by $\|e\|_a$ gives

$$\boxed{\|u - u_h\|_a \leq C_Q \sqrt{\frac{N_{\mathrm{ov}}}{k_0}} \, \eta.}$$

Thus $C_{\mathrm{rel}} = C_Q \sqrt{N_{\mathrm{ov}}/k_0}$. The constant is independent of the mesh size, although it depends on mesh shape regularity and the coefficient lower bound.

Reliability and efficiency answer different questions

Reliability is the global upper bound

$$\|e\|_a \leq C_{\mathrm{rel}}\eta.$$

It ensures that a small estimator implies a small energy error.
Local efficiency is a lower-bound statement of the form

$$\eta_e \leq C_{\mathrm{eff}} \left(\|e\|_{a,\omega_e} + \mathrm{osc}_e \right).$$

It ensures that a large local indicator is associated with error near $\mathcal{K}_e$, apart from the unresolved variation in the prescribed data. The quantity osc_e, called **data oscillation**, measures the part of f, q, or k that is not represented by the local polynomial data approximation used to evaluate the residual.
Today we proved reliability. Local efficiency requires a separate lower-bound argument and is not proved in this lecture.

Effectivity measures the scale of an estimator

When the exact error is available, define the **effectivity index**

$$I_{\text{eff}} = \frac{\eta}{\|u - u_h\|_a}.$$

An effectivity index near one means that η estimates the magnitude of the energy error accurately. A nearly constant value different from one means that the estimator tracks the error but consistently overestimates or underestimates it.
Effectivity is normally evaluated only in verification problems, because the exact error is unknown in the applications for which an estimator is needed.

These terms should not be interchanged:

reliability	the estimator bounds the error from above
efficiency	the indicator is not unnecessarily large
effectivity	the observed ratio of estimator to exact error

Summary and next step

- $R(u_h; v) = a(u - u_h, v)$ connects the computable residual functional to the unknown error.
- Elementwise integration by parts separates balance defects inside elements from flux defects on interior and Neumann facets.
- Galerkin orthogonality allows v to be replaced by $v - \mathcal{Q}_h v$; local approximation then supplies the powers of the local mesh size.
- The indicators η_e identify where the residual is concentrated, while reliability bounds the global energy error by η.

Exit question. For P_1 elements with elementwise-constant k and $f = 0$, can η still be nonzero? Identify the residual term that remains.

Next lecture. Use the local indicators to mark elements, refine the mesh conformingly, and repeat the solve–estimate–mark–refine loop.